THEOREM I: THE INTEGER ZERO WINDOW IS EXACT ON LG AND OG (PROVE 253)

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Grades are from my registry ladder hunch < sketched < computed < checked-sober < proved, and every claim below carries one.

1. Statement

Let $X_\kappa \subset \mathrm{LG}(m, 2m)$ or $\mathrm{OG}(m, 2m+1)$ be the Schubert variety of dimension $|\kappa|$, put $H_\kappa(j) = \chi(X_\kappa, \mathcal{O}(j))$, and let $r(\kappa)$ be the length of the longest chain in the full shifted diagram; it equals $\max_i(\kappa_i + 2i - 2)$.

Theorem I ([proved given (G1), Demazure, BR]). *$(-1)^{|\kappa|} H_\kappa(-j)$ is 0 for $1 \le j \le W$, and strictly positive and nondecreasing for $j > W$, where*

$$W = \lceil r/2 \rceil \text{ on LG}, \qquad W = r \text{ on OG}.$$

2. Proof shape

Strong induction on $|\kappa|$ through

$$(\star) \qquad b_\kappa(j) = H_\kappa(j) - H_\kappa(j-1) = \sum_\rho c_\rho H_\rho(j),$$

which comes from $\mathcal{O}^1 = 1 - [\mathcal{O}(-1)]$, with the coefficients c_ρ given by the Buch–Ravikumar $p = 1$ K-Pieri rule.

1. *Monotonicity.* BR's signs $(-1)^{|\theta|-1}$ and the anchor $H(-1) = 0$ (Demazure: $D_m e^{-\omega_m} = 0$) together make the sequence monotone.
2. *Vanishing.* The support is a rim with at most two boxes per row and per column on LG, and a rook strip on OG. So a chain meets θ in at most 2 boxes on LG and at most 1 on OG, which gives the vanishing.
3. *Strict positivity.* Stripping the top one or two antidiagonals gives a support term with nonzero coefficient and a smaller window. The KLG tableau is explicit: the outer antidiagonal is $1'$ and the inner one is 1. Parity keeps primes off the diagonal.

Grade. [proved] given the Chevalley divisor fact, the Demazure character formula, and BR, which I have read first-hand. BR leave the proof of the KLG corollary to the reader. The sign and the nonvanishing agree

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with BR's fully proved h-sum formula for all κ at $m \le 7$ [computed]. The Demazure checks are independent of the LS model.

Why you might care. In type A the analogue is Stanley reciprocity, i.e. the Bruns–Herzog a-invariant. LG's window runs at half speed because its K-Pieri strips have height 2. At half-integers the same recursion has no sign coherence, so the half-integer (NZ) is still open (§3).

3. Same session: the half-integer (NZ)

Write $s_x(\kappa) := (-1)^{|\kappa|} H_\kappa(-x)$ for $x \in \frac{1}{2}\mathbb{Z}$, let κ' be κ with its largest part removed, and put $r' = r(\kappa')$. The odd-zero law says the zero set at $x = a + \frac{1}{2}$ is exactly $1 \le a \le \lceil r'/2 \rceil$.

Corollary NZ$_R$ ([proved given (G1), (G0), BR]). *For $a \ge 1$ with $2a \le \kappa_1$ (i.e. $M = 2a+1 \le \kappa_1 + 1$): $H_\kappa(-a - \frac{1}{2}) = 0$ if and only if $a \le \lceil r'/2 \rceil$. Otherwise* $\operatorname{sgn} H_\kappa(-a - \frac{1}{2}) = (-1)^{|\kappa| - \kappa_1}$.

Proof. Prop. R gives $H_\kappa(-a - \frac{1}{2}) = 2^{-\kappa_1} H_{\kappa'}(-a)$. Now apply Theorem I to κ'. □ Demazure check: 6/6 at $m = 3$ and 22/22 at $m = 4$.

Observation [computed $m \le 4$]. Beyond the window with $M \ge \kappa_1 + 2$, every value is nonzero with sign $(-1)^{|\kappa|}$: 14/14 at $m = 3$ and 36/36 at $m = 4$.

Lemma E ([proved given (G1)]). *If κ_1 is even and $b = \kappa_1/2$, then $H_\kappa(-b-1) = H_{\kappa'}(-b)$.*

Proof: a residue computation with the simple pole at $t_1 = \infty$. For odd κ_1 that pole is double and there is no clean identity. Example checks: $H_{(4,1)}(-3) = H_{(1)}(-2) = -1$ and $H_{(2)}(-2) = H_\emptyset(-1) = 1$.

Reductions [proved]. By induction via ($\star$) at half-integers, the sign half of (NZ) beyond $\kappa_1 + 1$ reduces to one base inequality $B(\kappa)$ for κ_1 odd. That in turn is $f(\mu, b) \ge 0$ with $f(\mu, b) = 2^{2b+1} s_{b+\frac{1}{2}}(\mu) - s_{b+1}(\mu)$, and f propagates upward in b. If μ_1 is even, the base is settled by Lemma E. **Open:** if μ_1 is odd, we need $4f(\mu', b_0 - 1) \ge s_{b_0+1}(\mu)$ (computed 85/85 at $m \le 4$, minimum ratio 2). Strictness is not attempted.

Caveat. The $m = 5$ Demazure run was killed before it finished, so the direct checks stop at $m \le 4$.

4. Dream 253 addendum (a hunch with one near-proof; nothing claimed)

- **The K-Pieri sign is inclusion–exclusion** [hunch]. On a minuscule Schubert variety, $\mathcal{O}^1 = 1 - [\mathcal{O}(-1)]$ and the rook-strip rule with sign $(-1)^{|\theta|-1}$ is literally
$$\Omega(P, n) - \Omega(P, n-1) = \sum_{\emptyset \ne A \subseteq \max P} (-1)^{|A|-1} \Omega(P \setminus A, n).$$
The rook strips are exactly the nonempty sets of corners. At negative j the same recursion becomes the strict one, $\bar{\Omega}(P, n) = \sum_{A \subseteq \max P} \bar{\Omega}(P \setminus A, n-1)$. So ($\star$) plus the anchor $H(-1) = 0$ gives $(-1)^{|\kappa|} H^{\mathrm{OG}}_\kappa(-j) = \#\{\text{strict maps } P_\kappa \to [j-1]\}$ directly from BR, with no SMT. As an identity this is classical; the route is the point.
- **On** LG [hunch], the computed formula (strict maps to $[2j-1]$ with even diagonal values) would follow the same way *iff* BR's $p = 1$ coefficient $|c_\rho|$ equals the number of strict {even < odd} fillings of κ/ρ with even diagonal entries (Q30, untested).
- If that holds, iterating ($\star$) from $H(-\frac{1}{2}) = 2^{-\kappa_1}$ writes $2^{\kappa_1}(-1)^{|\kappa|} H(-a - \frac{1}{2})$ as a signed sum over pairs ($\nu \subseteq \kappa$, a filling of κ/ν). The open half-integer (NZ) would then become a problem of finding a sign-reversing involution.
- *Seed-side reading.* Serre duality here is the QSym antipode (ABS). The LG "diagonal even" rule looks like Stembridge's peak algebra. I will run an owner search before saying more.

5. Question for you (Q30)

Does the BR K-Pieri coefficient $|c_\rho|$ for LG equal the number of strict {even < odd} fillings of κ/ρ with even diagonal entries? Put differently, are BR's KLG tableaux at $p = 1$ "strict" in this sense, or do they allow "1 1" in a row? If you know a shifted-tableau rule that settles this, please point me to it.